

1. Administrative & Study Identification

**Full Study Title:** Extension Study for the Port Delivery System With Ranibizumab (Portal) **Short Title/Acronym:** Portal Study **Protocol Number:** GR40549 **NCT Number:** NCT03683251 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** Port Delivery System (PDS) with ranibizumab 100 mg/mL **Phase of Development:** Phase 3 **Trial Status:** Recruiting **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To evaluate the long-term safety and tolerability of the Port Delivery System (PDS) with ranibizumab 100 mg/mL given at 24-week intervals across the study duration in participants with neovascular age-related macular degeneration (nAMD). **Secondary Objectives:** To evaluate long-term efficacy, measured by the mean changes in best-corrected visual acuity (BCVA) and central subfield thickness (CST) from baseline over 240 weeks.

**2.2 Background & Rationale** To address the high treatment burden of frequent, standard-of-care intravitreal anti-VEGF injections in nAMD patients. The PDS is an innovative implant allowing continuous delivery of a customized formulation of ranibizumab into the vitreous. This study generates long-term data demonstrating that continuous delivery preserves vision and anatomical outcomes while drastically reducing the number of necessary clinical visits.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional (Open-label extension) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Rollover study) **Sub-studies:** \* **Sub-study 1 (Transscleral Photocoagulation):** Evaluates the effectiveness of using transscleral photocoagulation (TPC) with the Iridex laser system to mitigate vitreous hemorrhages secondary to the PDS implantation procedure (~55 participants). \* **Sub-study 2 (Re-implantation):** Evaluates the safety of re-implantation with the updated PDS with ranibizumab for patients who previously participated in the main study (up to 100 participants).

**3.2 Planned Enrollment & Duration Target \$N\$:** 1000 participants (Rollover participants from the Phase II Ladder, Phase III Archway, and Phase IIIb Velodrome trials, along with sub-study

enrollments) **Number of Sites:** Multicenter, global **Estimated Study Start:** 20-Sep-2018 **Estimated Primary Completion:** 31-Oct-2029

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Participants with a confirmed diagnosis of neovascular age-related macular degeneration (nAMD). 2. Successful completion of the Phase II Study GX28228 (Ladder), Phase III Study GR40548 (Archway), or Phase IIIb Study WR42221 (Velodrome), or completed Week 24 visit in Study WR42221 but were not eligible for randomization. 3. All subtypes of nAMD lesions are permissible (type I, II, III, or mixed) but must involve the macula (6 mm diameter centered at the fovea).

**4.2 Exclusion Criteria** 1. Pregnant or breastfeeding, or intending to become pregnant during the treatment period and for at least 28 days after the last ITV injection or 1 year after the last implant refill-exchange. 2. Any concurrent intraocular condition (e.g., cataract, glaucoma, diabetic retinopathy) that would require surgical intervention during the study to prevent or treat visual loss. 3. Active intraocular inflammation (grade trace or above), history of vitreous hemorrhage, or history of rhegmatogenous retinal detachment.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Port Delivery System (PDS) with ranibizumab (Trade Names: Lucentis, Susvimo, Cimerli, Byooviz) 100 mg/mL refilled at fixed 24-week intervals (Q24W). **Route of Administration:** Intravitreal implant (surgical placement with subsequent in-clinic refill-exchanges). **Duration of Treatment:** Up to 240 weeks (approximately 5 years).

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard concomitant systemic medications not interfering with ocular parameters. **Prohibited:** Concurrent use of systemic anti-VEGF agents and chronic use of oral corticosteroids.

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## 6. Study Timeline & Visit Schedule

| Visit ID     | Window (Days)  | Key Activities/Procedures                                                          |
|--------------|----------------|------------------------------------------------------------------------------------|
| Screening    | Roll-over      | Informed Consent, BCVA baseline assessment, SD-OCT imaging                         |
| Baseline     | Day 1          | PDS Implantation (if new/sub-study) or continuation of PDS, initial visual grading |
| Interim      | Every 24 Weeks | PDS refill-exchange procedures, Safety/AE assessment, BCVA & CST tests             |
| End of Study | Week 240+      | Final visual and anatomical assessments, PDS evaluation                            |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Incidence and severity of ocular and systemic adverse events (AEs), including adverse events of special interest (AESI) such as endophthalmitis, conjunctival bleb leak, and vitreous hemorrhage, over 240 weeks. **Measurement Method:** Clinical evaluation, slit-lamp biomicroscopy, and indirect ophthalmoscopy.

**7.2 Secondary & Exploratory Endpoints** \* Mean change in best-corrected visual acuity (BCVA) measured by Early Treatment Diabetic Retinopathy Study (ETDRS) letters from baseline over 240 weeks. \* Mean change in central subfield thickness (CST) from baseline over 240 weeks. \* Percentage of participants who lose <15, <10, or <5 letters in BCVA score from baseline over time.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Standard collection during visits, specifically tracking AESIs associated with the device placement and refill procedures (e.g., cataract, vitreous hemorrhage). **Serious Adverse Events (SAEs):** Evaluated continuously for device dislocation, severe infection, or unexpected vision loss, requiring 24-hour reporting. **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by Hoffmann-La Roche.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Safety Set encompasses all patients receiving the PDS via optimized surgery procedure. **Sample Size Justification:** As an extension trial, the sample size targets up to 1000 participants based on eligible rollover participants from the parent trials and distinct sub-study cohorts. **Handling of Missing Data:** Standard imputation (e.g., Last Observation Carried Forward [LOCF]) for missing continuous longitudinal data on visual outcomes.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Onsite clinical monitoring during key refill intervals; central reading center grading for FP and SD-OCT images. **Audit Trail:** eCRF locking and tracking of structural imaging adherence via standard quality audit mechanisms.

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## 11. Study Identifiers & Governance

**Primary ID:** GR40549 **Registry Number:** NCT03683251 **Posting ID:** 180205185033840651 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** Portal **Official Regulatory Title:** A Multicenter, Open-Label Extension Study to Evaluate the Long-Term Safety and Tolerability of the Port Delivery System With Ranibizumab in Patients With Neovascular Age-Related Macular Degeneration (Portal)

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## 12. Clinical Framework (nAMD Specific)

**Primary Condition:** Neovascular Age-Related Macular Degeneration (nAMD) **Diagnosis Threshold:** Previously diagnosed nAMD completing prior PDS trials **Medical Methodology:** Continuous intravitreal anti-VEGF delivery **Optimization Target:** Maintain BCVA and anatomical integrity with a reduced injection burden

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## 13. Study Procedures & Methodology

**Management Pathway:** Fixed Q24W clinical refill-exchange of the PDS implant **Education Protocol (HCP):** Specialized surgical training for implant placement and clinical training for the refill-exchange procedure **Education Protocol (Patient):** Post-operative care, eye protection, and endophthalmitis warning signs **Quality Audit Mechanism:** Central reading center validation for BCVA testing and OCT scans

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## 14. Observation & Follow-Up Schedule

**Total Duration:** Up to 240 weeks **Onsite Visit Cadence:** Every 24 weeks for refills and comprehensive evaluation **Remote Monitoring Frequency:** PRN (Pro re nata) based on patient-reported symptoms **Checklist Review Interval:** At every 24-week clinical visit

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## 15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|-----|------|-----|-----|
| Role                     | Name   | Signature | Date          | :-- | :--  | :-- | :-- |
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |     |      |     |     |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |     | Lead |     |     |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |     |      |     |     |